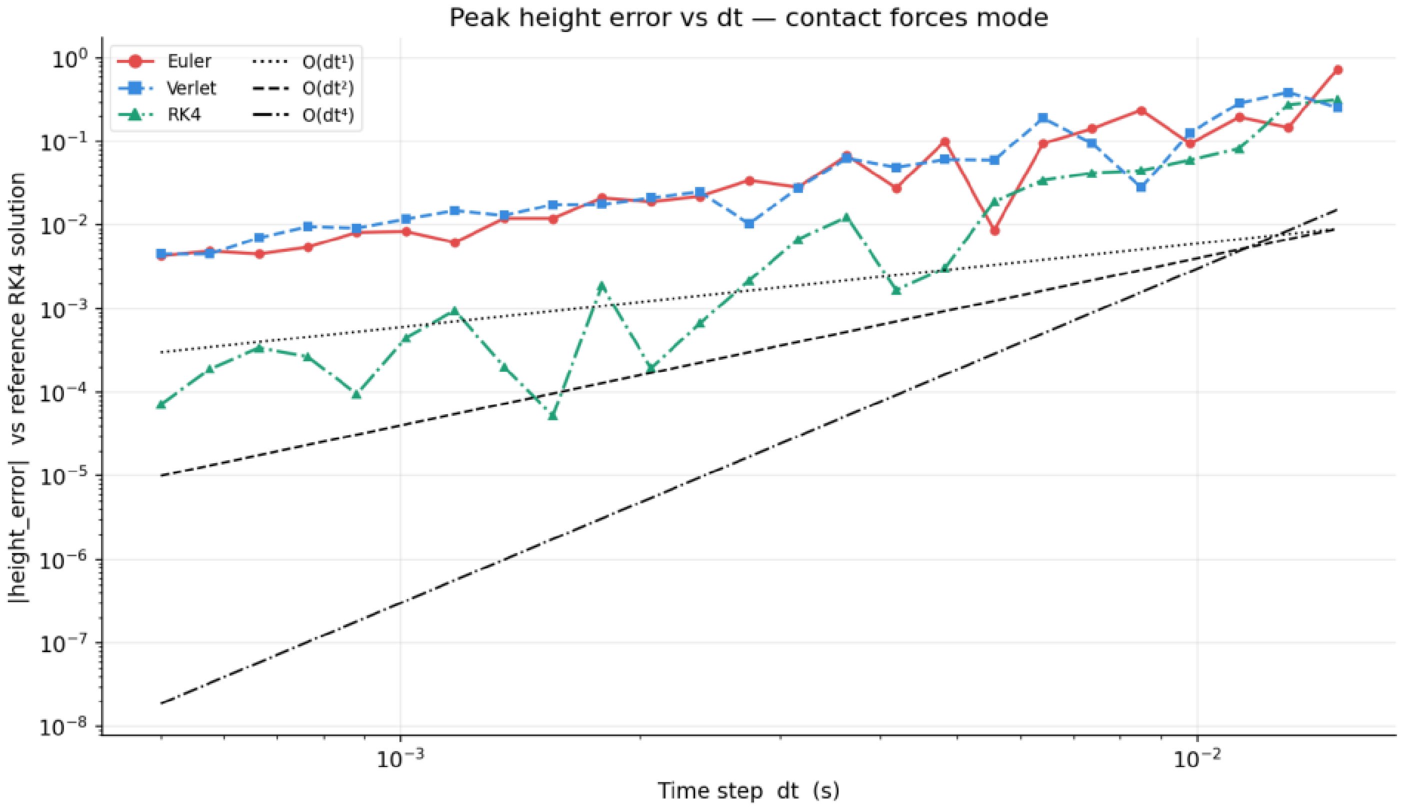


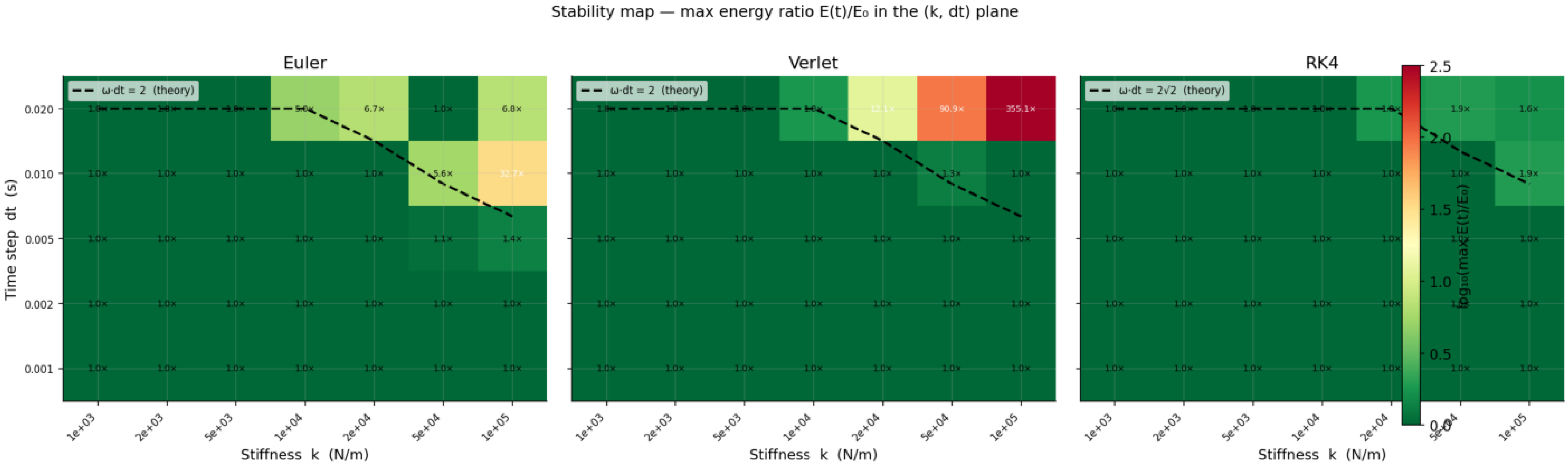

CONVERGENCE — CONTACT FORCES

Unlike impulse-based contact, the spring-damper force is continuous → integrator order is partially visible.

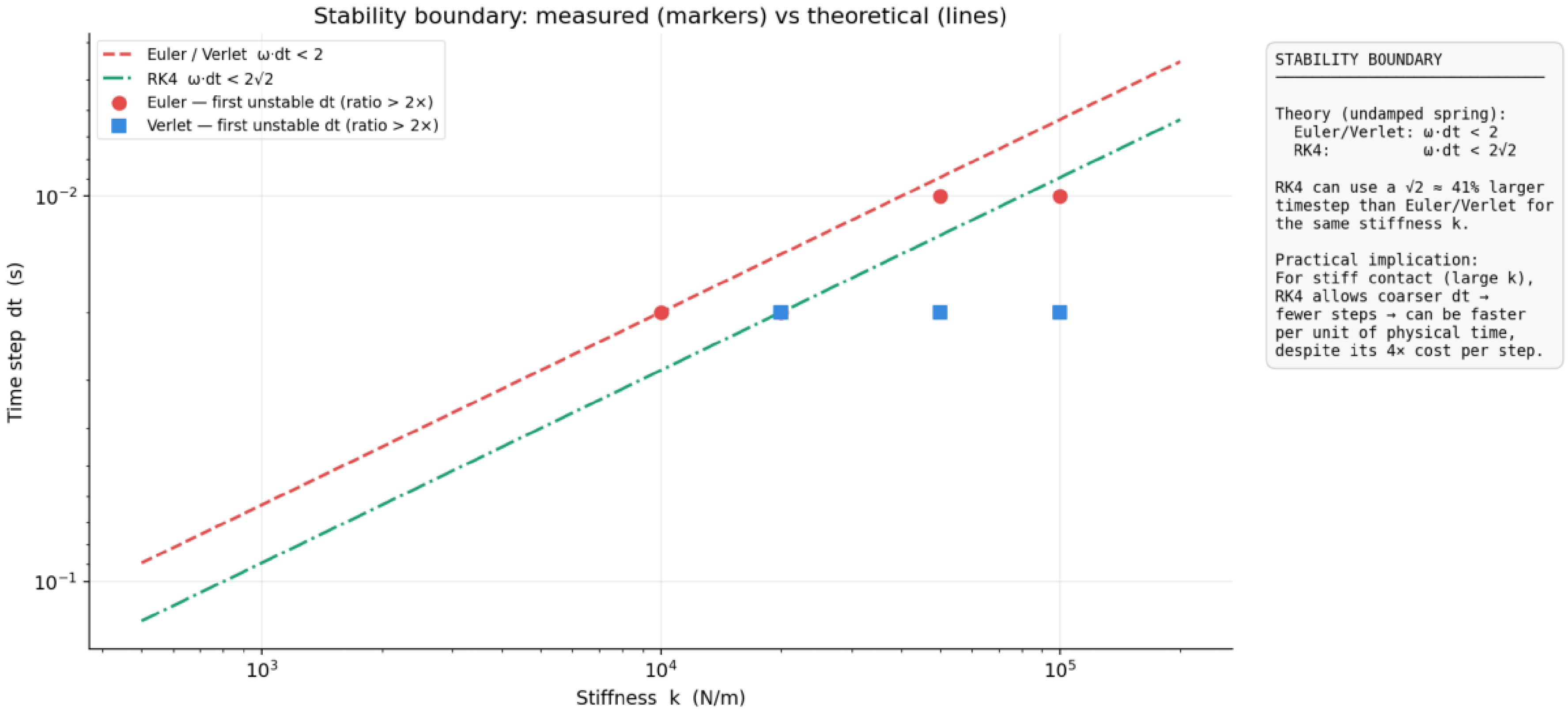
Slopes (measured below):
- Euler ≈ O(dt^1.3)
- Verlet ≈ O(dt^1.2)
- RK4 ≈ O(dt^2.4)

Why not $O(dt^2)$ / $O(dt^4)$?
The max(F_spring + F_damp, 0) clamp introduces a $C^0$ discontinuity at contact onset/offset, capping the effective global order.

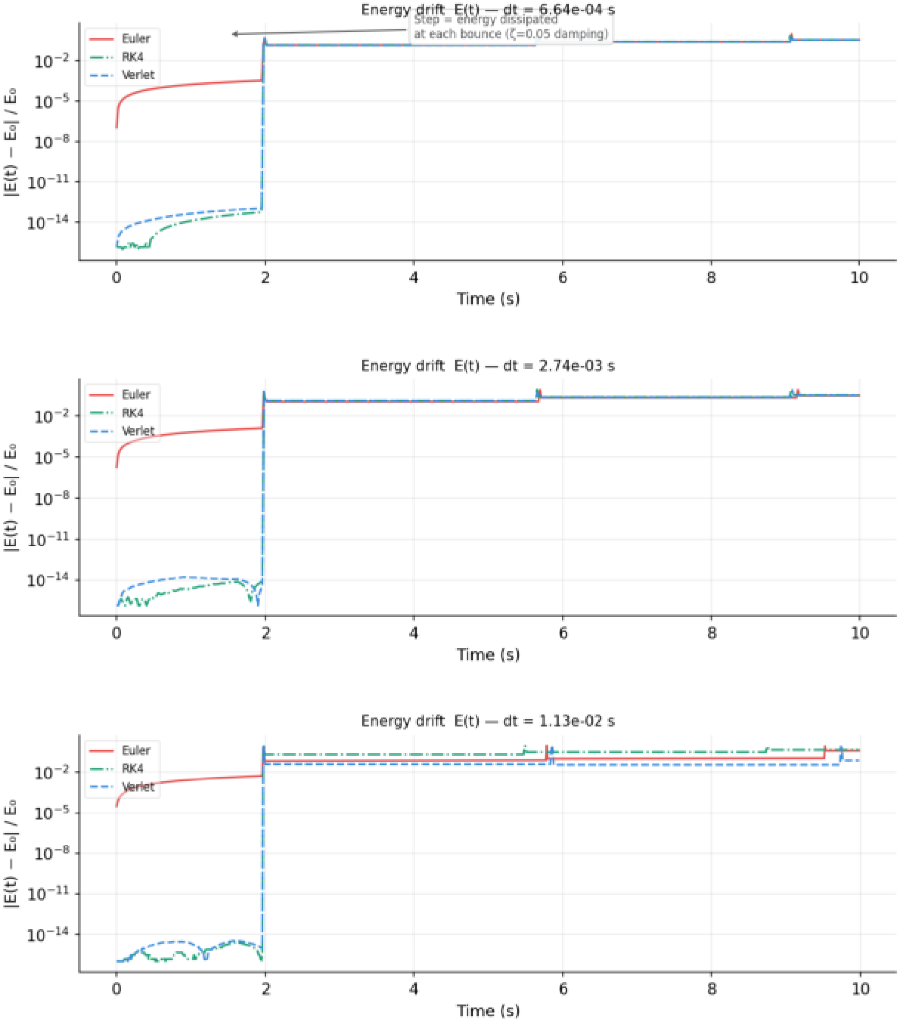
Despite this, RK4 is clearly the most accurate at any dt.



Stability map — max energy ratio E(t)/E₀ in the (k, dt) plane
Euler
Verlet
RK4
ω·dt = 2 (theory)
ω·dt = 2√2 (theory)
Time step dt (s)
Stiffness k (N/m)
log₁₀(max E(t)/E₀)



Stability boundary: measured (markers) vs theoretical (lines)
Euler / Verlet ω·dt < 2
RK4 ω·dt < 2√2
Euler — first unstable dt (ratio > 2×)
Verlet — first unstable dt (ratio > 2×)
Time step dt (s)
$10^{-2}$
$10^{-1}$
$10^{3}$
$10^{4}$
$10^{5}$
Stiffness k (N/m)
STABILITY BOUNDARY
Theory (undamped spring):
Euler/Verlet: ω·dt < 2
RK4: ω·dt < 2√2
RK4 can use a √2 ≈ 41% larger timestep than Euler/Verlet for the same stiffness k.
Practical implication:
For stiff contact (large k), RK4 allows coarser dt → fewer steps → can be faster per unit of physical time, despite its 4× cost per step.




ENERGY DRIFT E(t) — CONTACT FORCES

Staircase steps = energy lost at each bounce from viscous damping (ζ=0.05). This is physics, not numerical error.

Between bounces (free flight):
  Energy should be conserved.
  Euler drifts slowly (O(dt)).
  Verlet/RK4: near-exact.

The energy floor between steps is dt-dependent for Euler but solver-independent at the bounce steps (contact physics dominates there).